

Incident reconstruction

Neptune evidence pack

Pack: pack:sha256:b4ff87325ec9c9a6e08754220186a67ec95e7536fe1e702805d6ab65c564ddf2
Template: incident-timeline v1
(sha256:75a3744df6c388e3c93d2663197307ad8b94a0ffc20aed0f91cdd4950d560284)
Subject: event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
Interval: [1789394400000000000, 1789398000000000000) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416
Snapshot: snapshot:sha256:8031de7e9a50d529b998c084ed9c20fec559b2b6653e263394f8c22e7b04d4cb
Memory: graph-schema 1, Ledger head tx 5, generation
sha256:26982e7cc660cb2b8fdc9461ede70ee069fea13f16a04559e5a77db52596ca13, vocabulary 9
Inference: excluded (2 inferred claims left out)

One incident's timeline on the clock the pack interval chooses: the incident, the other records of it, events, interventions and diagnostics involving its machines, at its site or in its zone, and the statements of the documents that report it, each citing its evidence (Memory ADR 0013). An entry on another clock is placed only through a clock mapping Memory states, and names it; with none it is listed as not placed, never guessed. Records of one event that state different times are a conflict: every time is shown with its difference, none is chosen. Co-occurrence is listed apart and is never cause.

Every statement cites the claim ids it rests on. The claims index lists them; the appendices say how to resolve each evidence ref and record through the Ledger.

1. Records of this incident

Which other records Memory states are the same incident (same_as), and which it cannot decide (same_as_candidate: every reading, none chosen).

In scope:

event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14 - the pack subject

NOT COVERED - the snapshot holds no current claim of same_as, same_as_candidate about the nodes in scope within the interval.

2. Reconstruction on the chosen clock

Each placement on the pack clock in onset order: the event's stated kind, severity, description, machines, place and the record it rests on. A placement through a clock mapping names the mapping records. Records of one event are compared as one event.

In scope:

event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14 - via
claim:sha256:3970ed3cf7c3a4faf612eb39c8a80e40817b689cd9200c0fdab9cfff6719d4853,
claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77,
claim:sha256:dde86520c8e0b056ac3265ca450435ea655fcaa14989a9df56b0570e34425407
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/0 -
via claim:sha256:b3d4e1fea86e7396a3e39e2e498e570d5049a163ace841cd7a9e28a791dd8f92,
claim:sha256:e521488a22cdf82157f33185da3af789029c75e8fe94ab2c8ff07e457b16b05f
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/1 -
via claim:sha256:85911531522d52a7cbfadc980b0e29bcc163d9ef1c80716d1620d47c417d40e1,
claim:sha256:b3d4e1fea86e7396a3e39e2e498e570d5049a163ace841cd7a9e28a791dd8f92
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/2 -
via claim:sha256:b3d4e1fea86e7396a3e39e2e498e570d5049a163ace841cd7a9e28a791dd8f92,
claim:sha256:f42f5375a290ba43d010fa96c9837eddf02f8c593d3f186623640209d3eafecdc
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/3 -
via claim:sha256:421c929b6e15a5753f300047dcf7deb877eaf4c47d608d15a3e64285078e30f4,
claim:sha256:b3d4e1fea86e7396a3e39e2e498e570d5049a163ace841cd7a9e28a791dd8f92
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/4 -
via claim:sha256:b3d4e1fea86e7396a3e39e2e498e570d5049a163ace841cd7a9e28a791dd8f92,
claim:sha256:fb5505993a1670dcba22097ad2cfb3dad91eb325d288394886a07a78d4d65f64

[KNOWN] event

record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/0
valid [1789396080000000000, 1789396080000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416
- evidenced_by: record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
[stated]
cites claim:sha256:e521488a22cdf82157f33185da3af789029c75e8fe94ab2c8ff07e457b16b05f
- has_description: "Cell restarted after the break; PALLET_C3 started from the HMI" [stated]
cites claim:sha256:34e764af3218fe07cbe8e9408b5eadc99bb2c57a8c8f934463f106061c9be0c0

[KNOWN] event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14

valid [1789396320000000000, 1789396320000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416
- at_site: site site-code:PLANT-2 [stated]
cites claim:sha256:dde86520c8e0b056ac3265ca450435ea655fcaa14989a9df56b0570e34425407
- event_kind: "incident" [stated]

```

    cites claim:sha256:193721c937cedcecc85b2483dceac6d5f7da2b2a232ad1ee9bdd0eb3cb911ce3
- evidenced_by: record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
  [stated]
    cites claim:sha256:b3d4e1fea86e7396a3e39e2e498e570d5049a163ace841cd7a9e28a791dd8f92
- has_description: "During the pick at P1 the gripper fingers struck the locating edge of infeed
  fixture PF-3. Collision detection stopped the arm and the operator pressed the E-stop at OP-2.
  The part fell onto the floor guard. Nobody was inside the cell." [stated]
    cites claim:sha256:ba254fc1bac05fe8b5306b1b61be45cb675a4419f7acac87415c1bfff74b6a0b9
- in_zone: zone zone-code:CELL-3 [stated]
    cites claim:sha256:3970ed3cf7c3a4faf612eb39c8a80e40817b689cd9200c0fdab9cfff6719d4853
- involves: machine asset-tag:ARM-3A [stated]
    cites claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
- stated_severity: "Property damage, no injury" [stated]
    cites claim:sha256:f8f7db7f9e1f367bbf9cab98288e14bce328b175befb9c798d24c853718feb2a
[KNOWN] event
  record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/1
  valid [1789396358000000000, 1789396358000000001) ticks on
    rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416
- evidenced_by: record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
  [stated]
    cites claim:sha256:85911531522d52a7cbfadc980b0e29bcc163d9ef1c80716d1620d47c417d40e1
- has_description: "Collision detection on joint 5 at pick P1; protective stop" [stated]
    cites claim:sha256:c7848a1b700aa9ff268904b61d3b0eb855a725117aa4b04bf806cd253af5fb1a
[KNOWN] event
  record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/2
  valid [1789396361000000000, 1789396361000000001) ticks on
    rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416
- evidenced_by: record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
  [stated]
    cites claim:sha256:f42f5375a290ba43d010fa96c9837eddf02f8c593d3f186623640209d3eafecd
- has_description: "Operator presses the E-stop at OP-2" [stated]
    cites claim:sha256:d48901355f02f7fdb4d548188536db3f37704daab706a33f9f4c3c0b0533a1c
[KNOWN] event
  record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/3
  valid [1789396410000000000, 1789396410000000001) ticks on
    rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416
- evidenced_by: record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
  [stated]
    cites claim:sha256:421c929b6e15a5753f300047dcf7deb877eaf4c47d608d15a3e64285078e30f4
- has_description: "Cell supervisor notified; cell locked out" [stated]
    cites claim:sha256:74663942f9698a46f03935ebbd291f1f2724678b16e98a3da816d3ae9a28e5a
[KNOWN] event
  record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/4
  valid [1789397520000000000, 1789397520000000001) ticks on
    rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416
- evidenced_by: record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
  [stated]
    cites claim:sha256:fb5505993a1670dcba22097ad2cfb3dad91eb325d288394886a07a78d4d65f64
- has_description: "PF-3 locating edge found bent; part 7731-B dropped" [stated]
    cites claim:sha256:7b329d903f6ec0592718cfd16e356d86f1d883de32d853de10d4b9f72abd302

```

3. Configuration of the machines involved

The configuration chain of each machine the incident involves: decided, ambiguous or unknown spans, on their own clocks.

In scope:

```

machine asset-tag:ARM-3A - via
  claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77

```

On other clocks (never compared with the pack interval):

[KNOWN] machine asset-tag:ARM-3A

```

  valid [1772123400000000000, 1773171000000000000) ticks on
    rec:sha256:3c66443acb144517a1c805cdb5cd1649c33d69364a19e8cbfcc6ac87bf108cc2
- has_configuration: configuration cfg:cfg-c3-1.4 [stated]
    cites claim:sha256:7e90fd00cf552197c5f3167967fd6bc066af6a6ee4784ce14e0def4014a2178b

```

[UNKNOWN - stated as not known; the cited claim names the record leaving it open] machine

asset-tag:ARM-3A

```

  valid [1773171000000000000, 1787057700000000000) ticks on
    rec:sha256:3c66443acb144517a1c805cdb5cd1649c33d69364a19e8cbfcc6ac87bf108cc2
- configuration_unknown: record
    rec:sha256:1edfbeeaa4818e8773ee06a088e8f4bad464716ac405de6fa8e907fecdfb1a291 [stated] (unknown)

```

```
    reading)
    cites claim:sha256:fa7d8a46fe52aaace2491c7a75b4612136407eec673223e9497b1e0b4ffc9457
- configuration_unknown: record
    rec:sha256:cf0ff9b4c29e901cab9100bf2127dcf522569217235aee2172495ab59c20a44f [stated] (unknown
    reading)
    cites claim:sha256:54f3f0fbc3fd78c80887f62140cb845e1fb8615e3639440192279b876a0449d3
```

[KNOWN] machine asset-tag:ARM-3A

```
valid [17870577000000000000, 17890572000000000000) ticks on
    rec:sha256:3c66443acb144517a1c805cdb5cd1649c33d69364a19e8cbfcc6ac87bf108cc2
- has_configuration: configuration cfg:cfg-c3-1.5 [stated]
    cites claim:sha256:792ec98f356be97b64a334cc52a767e19271db6f2291d233d3892e3f657b9be7
```

[UNKNOWN - stated as not known; the cited claim names the record leaving it open] machine asset-tag:ARM-3A

```
valid [17890572000000000000, open) ticks on
    rec:sha256:3c66443acb144517a1c805cdb5cd1649c33d69364a19e8cbfcc6ac87bf108cc2
- configuration_unknown: record
    rec:sha256:e987ea4c6e44d868ca6e5c379698d30370df8543cf7af81bb4c6594f92e4b395 [stated] (unknown
    reading)
    cites claim:sha256:28a44a916613695062e47fe36ddb37f6a1692f3b83033300d0299bb6c3785767
- configuration_unknown: record
    rec:sha256:ec48abcf445fd73cf96bb11308219faaa723017c6eb94f30108c1d9432be9091 [stated] (unknown
    reading)
    cites claim:sha256:eb1a1e7f64da1dcc45d1af04b3f1a5d2b592d5c007d42dd18600f2946ac77a2b
```

4. Configuration changes

Each decided change on the involved machines' chains.

In scope:

```
configuration cfg:cfg-c3-1.4 - via
    claim:sha256:7e90fd00cf552197c5f3167967fd6bc066af6a6ee4784ce14e0def4014a2178b,
    claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
configuration cfg:cfg-c3-1.5 - via
    claim:sha256:792ec98f356be97b64a334cc52a767e19271db6f2291d233d3892e3f657b9be7,
    claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
```

NOT COVERED - the snapshot holds no current claim of succeeds about the nodes in scope within the interval.

5. Runs of the machines involved

The runs the involved machines recorded, with the configuration bound to each, on the run's own clock.

In scope:

```
run record:rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb - via
    claim:sha256:7da7b19c2e924f87cf549b458d08a63c660ea921c3de10bd220c840e6958207f,
    claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
run record:rec:sha256:b32f5af00eda27e4cdc6686b87cf19c87ca5a2799afd882df79bbd0005bb8a93 - via
    claim:sha256:62a5b611cb4b79db304fd777529ef988ee11ef2419d0b906676fcb090650dbb,
    claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
```

On other clocks (never compared with the pack interval):

```
[KNOWN] run record:rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb
    valid [17894105767000000000, 17894108857000000000) ticks on
    rec:sha256:e7535614127fb5c070e0a11cef5c1a5cc1eb040d715170a095e7d0ea1454401e
- configuration_active_during: configuration cfg:cfg-c3-1.5 [stated]
    cites claim:sha256:c9768e325d65e4ab3ab9de155f04f83d8b0616b4328c3adca220cd95d2741fe8
```

```
[KNOWN] run record:rec:sha256:b32f5af00eda27e4cdc6686b87cf19c87ca5a2799afd882df79bbd0005bb8a93
    valid [17889768956000000000, 17889772046000000000) ticks on
    rec:sha256:eb356a64d164b501f402ca2b5715ec6eed74023c53be5123c0d051b0d9f8a08e
- configuration_active_during: configuration cfg:cfg-c3-1.5 [stated]
    cites claim:sha256:55c3156dc047ebd3cfa3bd107f93fc1e46dcb801f53da5e3440e6d30984df71c
```

6. Clocks and mappings

The clocks the involved machines record on, and every mapping Memory states from them (its parameters as stated, never applied here). A clock no mapping relates to the pack clock is why an entry is not placed.

In scope:

```
clock rec:sha256:1a297c4c8c50f7bfb0fe5b10789c1ed1be5990d93239238e4239b850121b5c0f - via
    claim:sha256:74dc39dbd00af611017b7ab3ea5ff7f30d26a102d32508178dc67baad5d1c54e,
    claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
clock rec:sha256:e7535614127fb5c070e0a11cef5c1a5cc1eb040d715170a095e7d0ea1454401e - via
    claim:sha256:5002e33d4e2f8341d4377152a162b5b023bc73ca1e9ebba7f0bcb327fd66b4c,
    claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
clock rec:sha256:eb356a64d164b501f402ca2b5715ec6eed74023c53be5123c0d051b0d9f8a08e - via
    claim:sha256:54bf7048b78c22d28dcde3e01b4bb8e5409408febb4835c8bf8185e5743c7314,
```

```
claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
machine asset-tag:ARM-3A - via
claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
On other clocks (never compared with the pack interval):
[KNOWN] machine asset-tag:ARM-3A
valid [1789410480000000000, 1789410789000000001) ticks on
rec:sha256:1a297c4c8c50f7bfb0fe5b10789c1ed1be5990d93239238e4239b850121b5c0f
- has_clock: clock rec:sha256:1a297c4c8c50f7bfb0fe5b10789c1ed1be5990d93239238e4239b850121b5c0f
[observed]
cites claim:sha256:74dc39dbd00af611017b7ab3ea5ff7f30d26a102d32508178dc67baad5d1c54e
[KNOWN] machine asset-tag:ARM-3A
valid [1789410576700000000, 1789410885700000001) ticks on
rec:sha256:e7535614127fb5c070e0a11cef5c1a5cc1eb040d715170a095e7d0ea1454401e
- has_clock: clock rec:sha256:e7535614127fb5c070e0a11cef5c1a5cc1eb040d715170a095e7d0ea1454401e
[observed]
cites claim:sha256:5002e33d4e2f8341d43777152a162b5b023bc73ca1e9ebba70bcb327fd66b4c
[KNOWN] machine asset-tag:ARM-3A
valid [1788976895600000000, 1788977204600000001) ticks on
rec:sha256:eb356a64d164b501f402ca2b5715ec6eed74023c53be5123c0d051b0d9f8a08e
- has_clock: clock rec:sha256:eb356a64d164b501f402ca2b5715ec6eed74023c53be5123c0d051b0d9f8a08e
[observed]
cites claim:sha256:54bf7048b78c22d28cdce3e01b4bb8e5409408febb4835c8bf8185e5743c7314
Left out: 2 inferred claims (inference excluded).
```

7. Co-occurrence (never cause)

Pairs of events from different sources whose onsets fall within the configured window on one clock. Co-occurrence is not causation.

In scope:

```
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14 - via
claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
```

NOT COVERED - the snapshot holds no current claim of co_occurs_within about the nodes in scope within the interval.

Claims index

Every claim the pack cites, as the snapshot holds it (the JSON pack holds each in full).

```
claim:sha256:193721c937cedcecc85b2483dceac6d5f7da2b2a232ad1ee9bdd0eb3cb911ce3
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14 event_kind
"incident"
stated; valid [1789396320000000000, 1789396320000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
claim:sha256:28a44a916613695062e47fe36ddb37f6a1692f3b83033300d0299bb6c3785767
machine asset-tag:ARM-3A configuration_unknown record
rec:sha256:e987ea4c6e44d868ca6e5c379698d30370df8543cf7af81bb4c6594f92e4b395
stated; valid [1789057200000000000, open) ticks on
rec:sha256:3c66443acb144517a1c805cdb5cd1649c33d69364a19e8cbfccc6ac87bf108cc2; recorded at Ledger tx 2 (current);
1 evidence refs; records rec:sha256:e987ea4c6e44d868ca6e5c379698d30370df8543cf7af81bb4c6594f92e4b395
claim:sha256:34e764af3218fe07cbe8e9408b5eadc99bb2c57a8c8f934463f106061c9be0c0
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/0
has_description "Cell restarted after the break; PALLET_C3 started from the HMI"
stated; valid [1789396080000000000, 1789396080000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
claim:sha256:3970ed3cf7c3a4faf612eb39c8a80e40817b689cd9200c0fdab9cfff6719d4853
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14 in_zone zone
zone-code:CELL-3
stated; valid [1789396320000000000, 1789396320000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
claim:sha256:421c929b6e15a5753f300047dcf7deb877eaf4c47d608d15a3e64285078e30f4
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/3
evidenced_by record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
stated; valid [1789396410000000000, 1789396410000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
claim:sha256:5002e33d4e2f8341d43777152a162b5b023bc73ca1e9ebba70bcb327fd66b4c
machine asset-tag:ARM-3A has_clock clock
rec:sha256:e7535614127fb5c070e0a11cef5c1a5cc1eb040d715170a095e7d0ea1454401e
observed; valid [1789410576700000000, 1789410885700000001) ticks on
rec:sha256:e7535614127fb5c070e0a11cef5c1a5cc1eb040d715170a095e7d0ea1454401e; recorded at Ledger tx 3 (current);
1 evidence refs; records rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb
claim:sha256:54bf7048b78c22d28cdce3e01b4bb8e5409408febb4835c8bf8185e5743c7314
```

machine asset-tag:ARM-3A has_clock clock
rec:sha256:eb356a64d164b501f402ca2b5715ec6eed74023c53be5123c0d051b0d9f8a08e
observed; valid [1788976895600000000, 1788977204600000001) ticks on
rec:sha256:eb356a64d164b501f402ca2b5715ec6eed74023c53be5123c0d051b0d9f8a08e; recorded at Ledger tx 3 (current);
1 evidence refs; records rec:sha256:b32f5af00eda27e4cdc6686b87cf19c87ca5a2799afd882df79bbd0005bb8a93
claim:sha256:54f3f0fbc3fd78c80887f62140cb845e1fb8615e3639440192279b876a0449d3
machine asset-tag:ARM-3A configuration_unknown record
rec:sha256:cf0ff9b4c29e901cab9100bf2127dcf522569217235aee2172495ab59c20a44f
stated; valid [1773171000000000000, 1787057700000000000) ticks on
rec:sha256:3c66443acb144517a1c805cdb5cd1649c33d69364a19e8cbfcc6ac87bf108cc2; recorded at Ledger tx 2 (current);
1 evidence refs; records rec:sha256:cf0ff9b4c29e901cab9100bf2127dcf522569217235aee2172495ab59c20a44f
claim:sha256:55c3156dc047ebd3cfa3bd107f93fc1e46dc801f53da5e3440e6d30984df71c
run record:rec:sha256:b32f5af00eda27e4cdc6686b87cf19c87ca5a2799afd882df79bbd0005bb8a93
configuration_active_during configuration cfg:cfg-c3-1.5
stated; valid [1788976895600000000, 1788977204600000000) ticks on
rec:sha256:eb356a64d164b501f402ca2b5715ec6eed74023c53be5123c0d051b0d9f8a08e; recorded at Ledger tx 3 (current);
1 evidence refs; records rec:sha256:b32f5af00eda27e4cdc6686b87cf19c87ca5a2799afd882df79bbd0005bb8a93,
rec:sha256:d0afd33258a7e623032df753e5902a172f71ba5c9b08277fc79681fd966fe1e2
claim:sha256:62a5b611cb4b79db304fd777529ef988ee11ef2419d0b906676fcb090650dbb
run record:rec:sha256:b32f5af00eda27e4cdc6686b87cf19c87ca5a2799afd882df79bbd0005bb8a93 recorded_by
machine asset-tag:ARM-3A
stated; valid [1788976895600000000, 1788977204600000000) ticks on
rec:sha256:eb356a64d164b501f402ca2b5715ec6eed74023c53be5123c0d051b0d9f8a08e; recorded at Ledger tx 3 (current);
2 evidence refs; records rec:sha256:b32f5af00eda27e4cdc6686b87cf19c87ca5a2799afd882df79bbd0005bb8a93
claim:sha256:74663942f9698a46f03935ebdbf291f1f2724678b16e98a3da816d3ae9a28e5a
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/3
has_description "Cell supervisor notified; cell locked out"
stated; valid [1789396410000000000, 1789396410000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
claim:sha256:74dc39dbd00af611017b7ab3ea5ff7f30d26a102d32508178dc67baad5d1c54e
machine asset-tag:ARM-3A has_clock clock
rec:sha256:1a297c4c8c50f7bfb0fe5b10789c1ed1be5990d93239238e4239b850121b5c0f
observed; valid [1789410480000000000, 1789410789000000001) ticks on
rec:sha256:1a297c4c8c50f7bfb0fe5b10789c1ed1be5990d93239238e4239b850121b5c0f; recorded at Ledger tx 3 (current);
1 evidence refs; records rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb
claim:sha256:792ce98f356be97b64a334cc52a767e19271db6f2291d233d3892e3f657b9be7
machine asset-tag:ARM-3A has_configuration configuration cfg:cfg-c3-1.5
stated; valid [1787057700000000000, 1789057200000000000) ticks on
rec:sha256:3c66443acb144517a1c805cdb5cd1649c33d69364a19e8cbfcc6ac87bf108cc2; recorded at Ledger tx 2 (current);
2 evidence refs; records rec:sha256:89c8afcc8c8e8eaffa09e7ed054f7888aef9f7c07215c9b80db4ac4d76de3b8cc,
rec:sha256:e74d4fcee8b1e8cd13986cb3c1d8c31e3c7e2f93404b34e8f3b3e9c0afd5fa75
claim:sha256:7b329d903f6ec0592718cfd16e356d86f1d883de32d853de10d4b9f72abd302
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/4
has_description "PF-3 locating edge found bent; part 7731-B dropped"
stated; valid [1789397520000000000, 1789397520000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
claim:sha256:7da7b19c2e924f87cf549b458d08a63c660ea921c3de10bd220c840e6958207f
run record:rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb recorded_by
machine asset-tag:ARM-3A
stated; valid [1789410576700000000, 1789410885700000000) ticks on
rec:sha256:e7535614127fb5c070e0a11cef5c1a5cc1eb040d715170a095e7d0ea1454401e; recorded at Ledger tx 3 (current);
2 evidence refs; records rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb
claim:sha256:7e90fd00cf552197c5f3167967fd6bc066af6a6ee4784ce14e0def4014a2178b
machine asset-tag:ARM-3A has_configuration configuration cfg:cfg-c3-1.4
stated; valid [1772123400000000000, 1773171000000000000) ticks on
rec:sha256:3c66443acb144517a1c805cdb5cd1649c33d69364a19e8cbfcc6ac87bf108cc2; recorded at Ledger tx 2 (current);
1 evidence refs; records rec:sha256:8d948cbb15493c4495c645ed62ada6f628994e20ada0a4c7d8f0b18d0b9b0d16
claim:sha256:85911531522d52a7cbfad980b0e29bcc163d9ef1c80716d1620d47c417d40e1
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/1
evidenced_by record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
stated; valid [1789396358000000000, 1789396358000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14 involves
machine asset-tag:ARM-3A
stated; valid [1789396320000000000, 1789396320000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
claim:sha256:b3dd4e1fea86e7396a3e39e2e498e570d5049a163ace841cd7a9e28a791dd8f92
event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14 evidenced_by
record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
stated; valid [1789396320000000000, 1789396320000000001) ticks on
rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14

claim:sha256:ba254fc1bac05fe8b5306b1b61be45cb675a4419f7acac87415c1bfff74b6a0b9
 event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
 has_description "During the pick at P1 the gripper fingers struck the locating edge of infeed fixture PF-3. Collision detection stopped the arm and the operator pressed the E-stop at OP-2. The part fell onto the floor guard. Nobody was inside the cell."
 stated; valid [1789396320000000000, 1789396320000000001] ticks on
 rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
 1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14

claim:sha256:c7848a1b700aa9ff268904b61d3b0eb855a725117aa4b04bf806cd253af5fb1a
 event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/1
 has_description "Collision detection on joint 5 at pick P1; protective stop"
 stated; valid [1789396358000000000, 1789396358000000001] ticks on
 rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
 1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14

claim:sha256:c9768e325d65e4ab3ab9de155f04f83d8b0616b4328c3adca220cd95d2741fe8
 run record:rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb
 configuration_active_during configuration cfg:cfg-c3-1.5
 stated; valid [1789410576700000000, 1789410885700000000] ticks on
 rec:sha256:e7535614127fb5c070e0a11cef5c1a5cc1eb040d715170a095e7d0ea1454401e; recorded at Ledger tx 3 (current);
 1 evidence refs; records rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb,
 rec:sha256:123ddd28321e877366ee3a926eff04c76bccd1596c624fb26e2f7126ae197b90

claim:sha256:d48901355f02f7fdb4d548188536db3f37704daaab706a33f9f4c3c0b0533a1c
 event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/2
 has_description "Operator presses the E-stop at OP-2"
 stated; valid [1789396361000000000, 1789396361000000001] ticks on
 rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
 1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14

claim:sha256:dde86520c8e0b056ac3265ca450435ea655fcaa14989a9df56b0570e34425407
 event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14 at_site site
 site-code:PLANT-2
 stated; valid [1789396320000000000, 1789396320000000001] ticks on
 rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
 1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14

claim:sha256:e521488a22cdf82157f33185da3af789029c75e8fe94ab2c8ff07e457b16b05f
 event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/0
 evidenced_by record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
 stated; valid [1789396080000000000, 1789396080000000001] ticks on
 rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
 1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14

claim:sha256:eb1a1e7f64da1dcca45d1af04b3f1a5d2b592d5cc007d42dd18600f2946ac77a2b
 machine asset-tag:ARM-3A configuration_unknown record
 rec:sha256:ec48abcf445fd73cf96bb11308219faaa723017c6eb94f30108c1d9432be9091
 stated; valid [1789057200000000000, open] ticks on
 rec:sha256:3c66443acb144517a1c805cdb5cd1649c33d69364a19e8cbfcc6ac87bf108cc2; recorded at Ledger tx 2 (current);
 1 evidence refs; records rec:sha256:ec48abcf445fd73cf96bb11308219faaa723017c6eb94f30108c1d9432be9091

claim:sha256:f42f5375a290ba43d010fa96c9837eddf02f8c593d3f186623640209d3eafecdd
 event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/2
 evidenced_by record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
 stated; valid [1789396361000000000, 1789396361000000001] ticks on
 rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
 1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14

claim:sha256:f8f7db7f9e1f367bbf9cab98288e14bce328b175befb9c798d24c853718feb2a
 event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
 stated_severity "Property damage, no injury"
 stated; valid [1789396320000000000, 1789396320000000001] ticks on
 rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
 1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14

claim:sha256:fa7d8a46fe52aaace2491c7a75b4612136407eec673223e9497b1e0b4ffc9457
 machine asset-tag:ARM-3A configuration_unknown record
 rec:sha256:1edfbaaa4818e8773ee06a088e8f4bad464716ac405de6fa8e907fecdfb1a291
 stated; valid [1773171000000000000, 1787057700000000000] ticks on
 rec:sha256:3c66443acb144517a1c805cdb5cd1649c33d69364a19e8cbfcc6ac87bf108cc2; recorded at Ledger tx 2 (current);
 1 evidence refs; records rec:sha256:1edfbaaa4818e8773ee06a088e8f4bad464716ac405de6fa8e907fecdfb1a291

claim:sha256:fb5505993a1670dcb422097ad2cfb3dad91eb325d288394886a07a78d4d65f64
 event record:rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14/timeline/4
 evidenced_by record rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
 stated; valid [1789397520000000000, 1789397520000000001] ticks on
 rec:sha256:0fd804b59f80db99ff7275d65b5365b1438d07bbb26721b4bd65d2bb2f8ed416; recorded at Ledger tx 4 (current);
 1 evidence refs; records rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14

Appendix A - evidence references

Each evidence ref the cited claims rest on, and how to resolve it through the Ledger's catalog API at the snapshot's head.

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 claim:sha256:e521488a22cdf82157f33185da3af789029c75e8fe94ab2c8ff07e457b16b05f,
 claim:sha256:f42f5375a290ba43d010fa96c9837eddff02f8c593d3f186623640209d3eafecd,
 claim:sha256:f8f7db7f9e1f367bbf9cab98288e14bce328b175befb9c798d24c853718feb2a,
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 cited by claim:sha256:55c3156dc047ebd3cfa3bd107f93fc1e46dcb801f53da5e3440e6d30984df71c,
 claim:sha256:c9768e325d65e4ab3ab9de155f04f83d8b0616b4328c3adca220cd95d2741fe8
 Ledger catalog-api 1.6.0 resolve: {"as_of":5,"evidence_ref":{"locator":[{"kind":"json_pointer","pointer":"/config_revision"},"source":"sha256:57ea0f32041ae257fcb6bfed1c7c5a6738203f6290e7c0beccf3180a2245fb25"}}

E5. {"locator":[{"kind":"json_pointer","pointer":"/rosvbag2_bagfile_information"},"source":"sha256:13b72327894552ab66dd421ab7081c4ef2594e7b980ad77d48cd8b902c178c49"}
 cited by claim:sha256:54b7f048b78c22d28cdce3e01b4bb8e5409408febb4835c8bf8185e5743c7314
 Ledger catalog-api 1.6.0 resolve: {"as_of":5,"evidence_ref":{"locator":[{"kind":"json_pointer","pointer":"/rosvbag2_bagfile_information"},"source":"sha256:13b72327894552ab66dd421ab7081c4ef2594e7b980ad77d48cd8b902c178c49"}}

E6. {"locator":[{"kind":"json_pointer","pointer":"/rosvbag2_bagfile_information"},"source":"sha256:dc69a522a2f10db3e282bd0b1d0ab047ccf9eeb83c0fb139da2f456e16006c12"}
 cited by claim:sha256:5002e33d4e2f8341d43777152a162b5b023bc73ca1e9ebba70bcb327fd66b4c,
 claim:sha256:74dc39dbd00af611017b7ab3eaf5ff7f30d26a102d32508178dc67baad5d1c54e
 Ledger catalog-api 1.6.0 resolve: {"as_of":5,"evidence_ref":{"locator":[{"kind":"json_pointer","pointer":"/rosvbag2_bagfile_information"},"source":"sha256:dc69a522a2f10db3e282bd0b1d0ab047ccf9eeb83c0fb139da2f456e16006c12"}}

E7. {"locator":[{"kind":"json_pointer","pointer":"/rosvbag2_bagfile_information/starting_time"},"source":"sha256:13b72327894552ab66dd421ab7081c4ef2594e7b980ad77d48cd8b902c178c49"}
 cited by claim:sha256:62a5b611cb4b79db304fd777529ef988ee11ef2419d0b906676fcb090650dbb
 Ledger catalog-api 1.6.0 resolve: {"as_of":5,"evidence_ref":{"locator":[{"kind":"json_pointer","pointer":"/rosvbag2_bagfile_information/starting_time"},"source":"sha256:13b72327894552ab66dd421ab7081c4ef2594e7b980ad77d48cd8b902c178c49"}}

E8. {"locator":[{"kind":"json_pointer","pointer":"/rosvbag2_bagfile_information/starting_time"},"source":"sha256:dc69a522a2f10db3e282bd0b1d0ab047ccf9eeb83c0fb139da2f456e16006c12"}
 cited by claim:sha256:7da7b19c2e924f87cf549b458d08a63c660ea921c3de10bd220c840e6958207f
 Ledger catalog-api 1.6.0 resolve: {"as_of":5,"evidence_ref":{"locator":[{"kind":"json_pointer","pointer":"/rosvbag2_bagfile_information/starting_time"},"source":"sha256:dc69a522a2f10db3e282bd0b1d0ab047ccf9eeb83c0fb139da2f456e16006c12"}}

E9. {"locator":[{"kind":"json_pointer","pointer":"/runs/1"},"source":"sha256:1279658641566e0b752c487a4ff6ca85f6895996d94ed6fb9134572011382dba"}
 cited by claim:sha256:62a5b611cb4b79db304fd777529ef988ee11ef2419d0b906676fcb090650dbb
 Ledger catalog-api 1.6.0 resolve: {"as_of":5,"evidence_ref":{"locator":[{"kind":"json_pointer","pointer":"/runs/1"},"source":"sha256:1279658641566e0b752c487a4ff6ca85f6895996d94ed6fb9134572011382dba"}}

E10. {"locator":[{"kind":"json_pointer","pointer":"/runs/2"},"source":"sha256:1279658641566e0b752c487a4ff6ca85f6895996d94ed6fb9134572011382dba"}
 cited by claim:sha256:7da7b19c2e924f87cf549b458d08a63c660ea921c3de10bd220c840e6958207f
 Ledger catalog-api 1.6.0 resolve: {"as_of":5,"evidence_ref":{"locator":[{"kind":"json_pointer","pointer":"/runs/2"},"source":"sha256:1279658641566e0b752c487a4ff6ca85f6895996d94ed6fb9134572011382dba"}}

E11. {"locator":[{"kind":"row","row":1},"source":"sha256:f4e37e7d88632fff34f908769d4470480060cfb0e59942e6a24f95a24efdcfb3"}
 cited by claim:sha256:fa7d8a46fe52aaace2491c7a75b4612136407e6c673223e9497b1e0b4ffc9457
 Ledger catalog-api 1.6.0 resolve: {"as_of":5,"evidence_ref":{"locator":[{"kind":"row","row":1},"source":"sha256:f4e37e7d88632fff34f908769d4470480060cfb0e59942e6a24f95a24efdcfb3"}}

E12. {"locator":[{"kind":"row","row":1},"source":"sha256:fdca333962781db7f725a857dc6647b73e192666d2e9b505a2151e0502f1ef16"}
 cited by claim:sha256:54f3f0fbc3fd78c80887f62140cb845e1fb8615e3639440192279b876a0449d3
 Ledger catalog-api 1.6.0 resolve: {"as_of":5,"evidence_ref":{"locator":[{"kind":"row","row":1},"source":"sha256:fdca333962781db7f725a857dc6647b73e192666d2e9b505a2151e0502f1ef16"}}

E13. {"locator":[{"kind":"row","row":2},"source":"sha256:fdca333962781db7f725a857dc6647b73e192666d2e9b505a2151e0502f1ef16"}
 cited by claim:sha256:792ec98f356be97b64a334cc52a767e19271db6f2291d233d3892e3f657b9be7
 Ledger catalog-api 1.6.0 resolve: {"as_of":5,"evidence_ref":{"locator":[{"kind":"row","row":2},"source":"sha256:fdca333962781db7f725a857dc6647b73e192666d2e9b505a2151e0502f1ef16"}}

ca333962781db7f725a857dc6647b73e192666d2e9b505a2151e0502f1ef16"}}

E14. {"locator":[{"kind":"row", "row":3}], "source":"sha256:30be864d249a145a23a78fc8b38686e09a3ca1d3c8b4c9f08c08feb847a3f2be"}
cited by claim:sha256:792ec98f356be97b64a334cc52a767e19271db6f2291d233d3892e3f657b9be7
Ledger catalog-api 1.6.0 resolve: {"as_of":5, "evidence_ref":{"locator":[{"kind":"row", "row":3}], "source":"sha256:30be864d249a145a23a78fc8b38686e09a3ca1d3c8b4c9f08c08feb847a3f2be"}}

E15. {"locator":[{"kind":"row", "row":6}], "source":"sha256:f4e37e7d88632fff34f908769d4470480060cfb0e59942e6a24f95a24efdcfb3"}
cited by claim:sha256:eb1a1e7f64da1dcc45d1af04b3f1a5d2b592d5c007d4d2dd18600f2946ac77a2b
Ledger catalog-api 1.6.0 resolve: {"as_of":5, "evidence_ref":{"locator":[{"kind":"row", "row":6}], "source":"sha256:f4e37e7d88632fff34f908769d4470480060cfb0e59942e6a24f95a24efdcfb3"}}

E16. {"locator":[{"kind":"row", "row":7}], "source":"sha256:f4e37e7d88632fff34f908769d4470480060cfb0e59942e6a24f95a24efdcfb3"}
cited by claim:sha256:28a44a916613695062e47fe36ddb37f6a1692f3b83033300d0299bb6c3785767
Ledger catalog-api 1.6.0 resolve: {"as_of":5, "evidence_ref":{"locator":[{"kind":"row", "row":7}], "source":"sha256:f4e37e7d88632fff34f908769d4470480060cfb0e59942e6a24f95a24efdcfb3"}}

Appendix B - records

R1. rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb
cited by claim:sha256:5002e33d4e2f8341d43777152a162b5b023bc73ca1e9ebba7f0bcb327fd66b4c,
claim:sha256:74dc39dbd00af611017b7ab3ea5ff7f30d26a102d32508178dc67baad5d1c54e,
claim:sha256:7da7b19c2e924f87cf549b458d08a63c660ea921c3de10bd220c840e6958207f,
claim:sha256:c9768e325d65e4ab3ab9de155f04f83d8b0616b4328c3adca220cd95d2741fe8
Ledger catalog-api 1.6.0 threads_of:
{"as_of":5, "record_id":"rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb"}
Ledger catalog-api 1.6.0 lineage:
{"as_of":5, "record_id":"rec:sha256:0573b8eb2f79453532a08126a34d064205827e2df34d1530342d4f4f12ff39bb"}

R2. rec:sha256:123ddd28321e877366ee3a926eff04c76bccd1596c624fb26e2f7126ae197b90
cited by claim:sha256:c9768e325d65e4ab3ab9de155f04f83d8b0616b4328c3adca220cd95d2741fe8
Ledger catalog-api 1.6.0 threads_of:
{"as_of":5, "record_id":"rec:sha256:123ddd28321e877366ee3a926eff04c76bccd1596c624fb26e2f7126ae197b90"}
Ledger catalog-api 1.6.0 lineage:
{"as_of":5, "record_id":"rec:sha256:123ddd28321e877366ee3a926eff04c76bccd1596c624fb26e2f7126ae197b90"}

R3. rec:sha256:1edfbeaa4818e8773ee06a088e8f4bad464716ac405de6fa8e907fecdfb1a291
cited by claim:sha256:fa7d8a46fe52aaace2491c7a75b4612136407eec673223e9497b1e0b4ffc9457
Ledger catalog-api 1.6.0 threads_of:
{"as_of":5, "record_id":"rec:sha256:1edfbeaa4818e8773ee06a088e8f4bad464716ac405de6fa8e907fecdfb1a291"}
Ledger catalog-api 1.6.0 lineage:
{"as_of":5, "record_id":"rec:sha256:1edfbeaa4818e8773ee06a088e8f4bad464716ac405de6fa8e907fecdfb1a291"}

R4. rec:sha256:89c8af8c8e8eaffa09e7ed054f7888aef9f7c07215c9b80db4acad76de3b8cc
cited by claim:sha256:792ec98f356be97b64a334cc52a767e19271db6f2291d233d3892e3f657b9be7
Ledger catalog-api 1.6.0 threads_of:
{"as_of":5, "record_id":"rec:sha256:89c8af8c8e8eaffa09e7ed054f7888aef9f7c07215c9b80db4acad76de3b8cc"}
Ledger catalog-api 1.6.0 lineage:
{"as_of":5, "record_id":"rec:sha256:89c8af8c8e8eaffa09e7ed054f7888aef9f7c07215c9b80db4acad76de3b8cc"}

R5. rec:sha256:8d948cbb15493c4495c645ed62ada6f628994e20ada0a4c7d8f0b18d0b9b0d16
cited by claim:sha256:7e90fd00cf552197c5f3167967fd6bc066af6a6ee4784ce14e0def4014a2178b
Ledger catalog-api 1.6.0 threads_of:
{"as_of":5, "record_id":"rec:sha256:8d948cbb15493c4495c645ed62ada6f628994e20ada0a4c7d8f0b18d0b9b0d16"}
Ledger catalog-api 1.6.0 lineage:
{"as_of":5, "record_id":"rec:sha256:8d948cbb15493c4495c645ed62ada6f628994e20ada0a4c7d8f0b18d0b9b0d16"}

R6. rec:sha256:b32f5af00eda27e4cdc6686b87cf19c87ca5a2799afd882df79bbd0005bb8a93
cited by claim:sha256:54b7f048b78c22d28cdce3e01b4bb8e5409408febb4835c8bf8185e5743c7314,
claim:sha256:55c3156dc047ebd3cfa3bd107f93fc1e46dc801f53da5e3440e6d30984df71c,
claim:sha256:62a5b611cb4b79db304fd777529ef988ee11ef2419d0b906676fcb090650dbb
Ledger catalog-api 1.6.0 threads_of:
{"as_of":5, "record_id":"rec:sha256:b32f5af00eda27e4cdc6686b87cf19c87ca5a2799afd882df79bbd0005bb8a93"}
Ledger catalog-api 1.6.0 lineage:
{"as_of":5, "record_id":"rec:sha256:b32f5af00eda27e4cdc6686b87cf19c87ca5a2799afd882df79bbd0005bb8a93"}

R7. rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14
cited by claim:sha256:193721c937cedcecc85b2483dceac6d5f7da2b2a232ad1ee9bdd0eb3cb911ce3,
claim:sha256:34e764af3218fe07cbe8e9408b5eadc99bb2c57a8c8f934463f106061c9be0c0,
claim:sha256:3970ed3cf7c3a4faf612eb39c8a80e40817b689cd9200c0fdab9cfff6719d4853,
claim:sha256:421c929b61e5a5753f300047dcf7deb877eaf4c47d608d15a3e64285078e30f4,
claim:sha256:74663942f9698a46f03935ebddf291f1f2724678b16e98a3da816d3ae9a28e5a,
claim:sha256:7b329d903f6ec0592718cfd16e356d86f1d883de32d853de10d4b9f72abd302,
claim:sha256:85911531522d52a7cbfad980b0e29bcc163d9ef1c80716d1620d47c417d40e1,
claim:sha256:86f293e1fe519c2e8f372d5e4606b607b648da54e3482ad3f11d7b97e0e32b77,
claim:sha256:b3d4e1fea86e7396a3e39e2e498e570d5049a163ace841cd7a9e28a791dd8f92,
claim:sha256:ba254fc1bac05fe8b5306b1b61be45cb675a4419f7acac87415c1bff74b6a0b9,
claim:sha256:c7848a1b700aa9ff268904b61d3b0eb855a725117aa4b04bf806cd253af5fb1a,
claim:sha256:d48901355f02f7fdb4d548188536db3f37704daaab706a33f9f4c3c0b0533a1c,
claim:sha256:dde86520c8e0b056ac3265ca450435ea655fcaa14989a9df56b0570e34425407,
claim:sha256:e521488a22cdf82157f3185da3af789029c75e8fe94ab2c8ff07e457b16b05f,
claim:sha256:f42f5375a290ba43d010fa96c9837eddf02f8c593d3f186623640209d3eafecd,
claim:sha256:f8f7db7f9e1f367bbf9cab98288e14bce328b175befb9c798d24c853718feb2a,
claim:sha256:fb5505993a1670dcba22097ad2cfc3dad91eb325d288394886a07a78d4d65f64
Ledger catalog-api 1.6.0 threads_of:
{"as_of":5, "record_id":"rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14"}

Ledger catalog-api 1.6.0 lineage:
 {"as_of":5,"record_id":"rec:sha256:c82df9cc3d8a3a5963d028fbc6906eca4450da73211dcb1cf2d55022f9664a14"}

R8. rec:sha256:cf0ff9b4c29e901cab9100bf2127dcf522569217235aee2172495ab59c20a44f
 cited by claim:sha256:54f3f0fbc3fd78c80887f62140cb845e1fb8615e3639440192279b876a0449d3
 Ledger catalog-api 1.6.0 threads_of:
 {"as_of":5,"record_id":"rec:sha256:cf0ff9b4c29e901cab9100bf2127dcf522569217235aee2172495ab59c20a44f"}

Ledger catalog-api 1.6.0 lineage:
 {"as_of":5,"record_id":"rec:sha256:cf0ff9b4c29e901cab9100bf2127dcf522569217235aee2172495ab59c20a44f"}

R9. rec:sha256:d0afd33258a7e623032df753e5902a172f71ba5c9b08277fc79681fd966fe1e2
 cited by claim:sha256:55c3156dc047ebd3cfa3bd107f93fc1e46dcb801f53da5e3440e6d30984df71c
 Ledger catalog-api 1.6.0 threads_of:
 {"as_of":5,"record_id":"rec:sha256:d0afd33258a7e623032df753e5902a172f71ba5c9b08277fc79681fd966fe1e2"}

Ledger catalog-api 1.6.0 lineage:
 {"as_of":5,"record_id":"rec:sha256:d0afd33258a7e623032df753e5902a172f71ba5c9b08277fc79681fd966fe1e2"}

R10. rec:sha256:e74d4feee8b1e8cd13986cb3c1d8c31e3c7e2f93404b34e8f3b3e9c0afd5fa75
 cited by claim:sha256:792ec98f356be97b64a334cc52a767e19271db6f2291d233d3892e3f657b9be7
 Ledger catalog-api 1.6.0 threads_of:
 {"as_of":5,"record_id":"rec:sha256:e74d4feee8b1e8cd13986cb3c1d8c31e3c7e2f93404b34e8f3b3e9c0afd5fa75"}

Ledger catalog-api 1.6.0 lineage:
 {"as_of":5,"record_id":"rec:sha256:e74d4feee8b1e8cd13986cb3c1d8c31e3c7e2f93404b34e8f3b3e9c0afd5fa75"}

R11. rec:sha256:e987ea4c6e44d868ca6e5c379698d30370df8543cf7af81bb4c6594f92e4b395
 cited by claim:sha256:28a44a916613695062e47fe36ddb37f6a1692f3b83033300d0299bb6c3785767
 Ledger catalog-api 1.6.0 threads_of:
 {"as_of":5,"record_id":"rec:sha256:e987ea4c6e44d868ca6e5c379698d30370df8543cf7af81bb4c6594f92e4b395"}

Ledger catalog-api 1.6.0 lineage:
 {"as_of":5,"record_id":"rec:sha256:e987ea4c6e44d868ca6e5c379698d30370df8543cf7af81bb4c6594f92e4b395"}

R12. rec:sha256:ec48abcf445fd73cf96bb11308219faaa723017c6eb94f30108c1d9432be9091
 cited by claim:sha256:eb1a1e7f64da1dcc45d1af04b3f1a5d2b592d5c007d42dd18600f2946ac77a2b
 Ledger catalog-api 1.6.0 threads_of:
 {"as_of":5,"record_id":"rec:sha256:ec48abcf445fd73cf96bb11308219faaa723017c6eb94f30108c1d9432be9091"}

Ledger catalog-api 1.6.0 lineage:
 {"as_of":5,"record_id":"rec:sha256:ec48abcf445fd73cf96bb11308219faaa723017c6eb94f30108c1d9432be9091"}